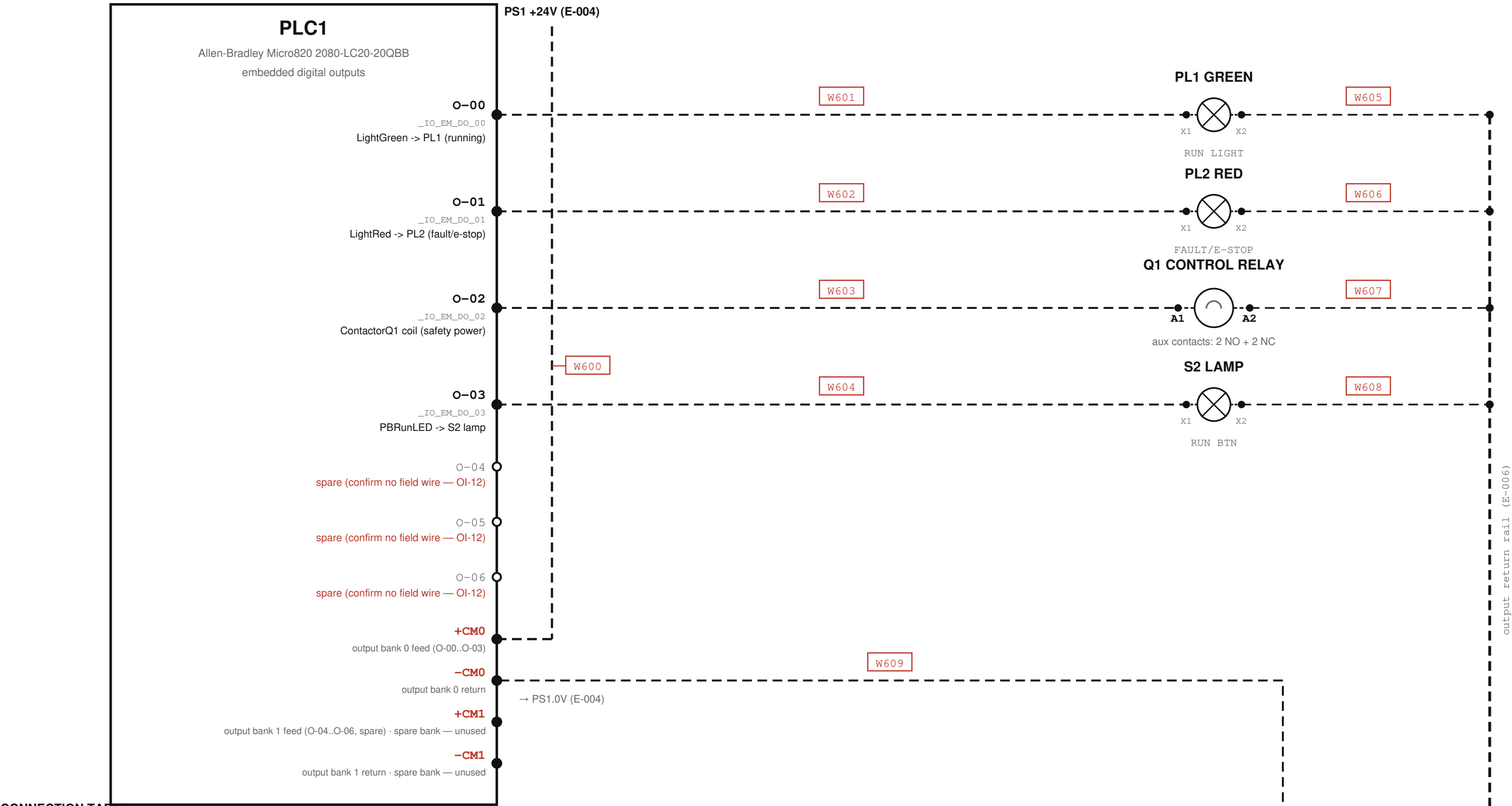


E-006 PLC OUTPUTS

Conv\_Simple / CV-101 (garage conveyor) — O-00..O-06 · pilots, Q1/MLC control-relay coil, button lamp · GS10 control = HYBRID Modbus + hardwired I/O (E-007; map OI-22)



CONNECTION TABLE

| Wire | From      | To                         | Signal                          | Type           | Status       | Notes |
|------|-----------|----------------------------|---------------------------------|----------------|--------------|-------|
| W600 | PS1.+24V  | PLC1.+CM0                  | output bank 0 feed              | power_24vdc    | field_verify |       |
| W601 | PLC1.O-00 | PL1.X1                     | LightGreen (run light)          | control_output | field_verify |       |
| W602 | PLC1.O-01 | PL2.X1                     | LightRed (fault/e-stop light)   | control_output | field_verify |       |
| W603 | PLC1.O-02 | Q1.A1                      | ContactorQ1 coil (safety power) | control_output | field_verify |       |
| W604 | PLC1.O-03 | S2.X1                      | PBRunLED (run button lamp)      | control_output | field_verify |       |
| W605 | PL1.X2    | output return rail (E-006) | run light return                | control_return | field_verify |       |
| W606 | PL2.X2    | output return rail (E-006) | fault light return              | control_return | field_verify |       |
| W607 | Q1.A2     | output return rail (E-006) | contactor coil return           | control_return | field_verify |       |
| W608 | S2.X2     | output return rail (E-006) | PB lamp return                  | control_return | field_verify |       |
| W609 | PLC1.-CM0 | PS1.0V                     | output bank 0 return            | power_0v       | field_verify |       |

Hardwired fallback documented in MIRA-WI-001 v3 §3 Fig 1 (DO\_03..DO\_07 → GS10 DI1..DI5 + DCM) — NOT ACTIVE and NOT drawn: P02.0x at factory default per the 2026-05-20 parameter export; DO\_07 does not exist on the 2080-LC20-20QBB (O-00..O-06); DO\_03 collides with PBRunLED in the live I/O map. Bench presence of any such wiring = FIELD VERIFY (OI-18).

SAFETY

- Monitored outputs are NOT a safety function.
- E-stop must remove power per NFPA 79.
- Q1/MLC is a CONTROL RELAY (24 VDC coil, driven by O-02) whose 2 NO contacts (13-14, 43-44) switch the drive's 230 V 1φ supply (V6 correction, E-003) — de-energizing O-02 DOES remove Q1's switched supply to the GS10. Q1 is NOT an NFPA-79/EN-60204-1 safety-rated disconnect (small control-relay contacts, not a lockable safety contactor); the e-stop (S0) → O-02 relationship is FIELD VERIFY (OI-26). Do not treat this circuit as the LOTO isolation point — always LOTO at CB1 or upstream.
- O-02 is the sole MLC/Q1 drive-enable coil output — do not reuse it for any other load (MIRA\_PLC\_WorkInstruction\_v3.pdf §3: "DO\_02 is already the MLC drive-enable coil and must never be reused").

NOTES

- Q1/MLC = Schneider TeSys CA3KN22BD control relay (2 NO + 2 NC, 24 VDC coil); coil A1(+)/A2(-) driven by O-02 (verified from photo wire\_3). NO contacts 13-14/43-44 switch the VFD's 230 V 1φ supply (see E-003); 21-22/31-32 NC unused.
- GS10 control (full-res photo): run/dir/speed = Modbus — top FWD/REV/DI connector is EMPTY, RJ45 landed. Bottom analog/output row (+10V/ACM/AI/AO1/DO1/DOC) is wired — likely speed reference + feedback. Map = OI-22.
- NO contacts 13-14 & 43-44 switch the 230 V 1φ VFD supply (see E-003); coil A1/A2 from O-02. NC 21-22/31-32 unused.
- GS10 run/dir/freq = Modbus — see E-007 (P00.20=1, P00.21=2 per 2026-05-20 parameter export)
- No as-built: O-00/O-01/O-03 from CCW v4.0; O-02 corroborated by live Prog\_init v2.1.
- Output bank technology + common feed = FIELD VERIFY (OI-09); WI-001 p.4 says relay dry contacts; ±CM polarity naming suggests DC-fed banks — conflicting evidence, resolve per 2080-IN009 + meter.
- GS10 manual warns against cycling a power-circuit switching device for normal run/stop — emergency situations only (GS10\_UM.txt L1754-1757, L1811-1813). This applies to Q1/MLC (energized by O-02, switches the drive's 230 V 1φ supply — see E-003): O-02/Q1 stays energized during normal operation; run/stop is commanded via Modbus (E-007), never by cycling Q1.
- S2 lamp (this sheet, O-03/PBRunLED) and S2's run/rearm pushbutton contact (E-005, I-04) are two elements of the SAME physical device — same cross-reference symmetry as Q1/MLC's coil (this sheet) ↔ power contacts (E-003).

SOURCES:

CCW\_VARIABLES\_v4.0.txt  
Prog\_init\_ConvSimple\_v2.1.st  
MIRA\_PLC\_WorkInstruction\_v3.pdf  
GS10\_actual\_parameters\_5.20.26.xlsx  
GS10\_UM.txt  
photos review/photos/wire\_1..4.jpg (2026-07-11, actual devices)  
full-res photos review/photos/"\_full.jpg + crops (2026-07-11);  
review/PHOTO\_EVIDENCE\_V5.md

LEGEND

- VERIFIED (solid)
- - - FIELD VERIFY (dashed)
- Wxxx proposed wire number (red = unverified)

PLC OUTPUTS

Conv\_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM

output map CCW v4.0 + live Prog\_init v2.1; O-02 do-not-reuse (MIRA\_PLC\_WorkInstruction\_v3.pdf §3) — see SAFETY

SHEET

E-006

REV A

2026-07-11

6 of 9